

Digital Waste Management:
Using a digital scale to build sustainable waste management habits

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Abstract

In this report we design and prototype a digital scale to help people build more sustainable waste management habits. Our project's purpose is to provide timely, concrete, and useful feedback on the impacts of changes in the user's waste patterns. Our prototype is powered by an Arduino Uno which reads a signal from stress/strain sensors and outputs data to an SD card, with input from a rudimentary user interface. Through testing we were able to successfully measure, log, and graphically analyze waste disposal over time. Future steps include further testing, greater simplicity and reliability, improved user interface, and IoT implementation.

Introduction

One of the greatest problems facing humanity today is the creation of extraordinary amounts of municipal solid waste (MSW), over 4.5 trillion pounds annually worldwide¹. This directly contributes to two of the greatest existential threats that currently face the planet: Global warming due to an accelerated greenhouse effect, and the human pollution that is destroying countless ecosystems and throwing the biodiversity of the planet into jeopardy. This problem is not one that can be solved by a machine made by three high school students for under \$150. It will require myriad dedicated specialists and scientists as well as the cooperation of a large percentage of the world's population in altering habits formed over centuries, in addition to the money and political influence to combat the millions who do not see our current trajectory as a problem.

Here we grapple with a small section of this broad issue: What can an individual do in their own home to decrease their personal impact? Much of this problem, particularly in the United States, comes from wasteful personal habits. In order to help combat these habits, people need to be able to see the tangible impact of changes they make. Thus, a system is required to measure their improvement. To address this, we have created a digital scale system that is capable of recording and logging weight measurements of trash at regular intervals. This is a two part system consisting of the weighing platform which houses the stress/strain sensors and the control box which interprets the signals from the sensors and user input, then logs the data. The prototype design has enough functionality to be a better alternative to DIY methods of measuring

¹ Clifford, C. (2021, May 30). *Trillions of pounds of trash: New Technology tries to solve an old garbage problem*. CNBC.
<https://www.cnn.com/2021/05/29/can-new-technology-solve-a-trillion-pound-garbage-problem.html#:~:text=Humans%20generate%20a%20remarkable%20amount,3.4%20billion%20tonnes%20by%202050>.

trash but lacks a competitive price point or polished design, something that might be solved in the future by taking advantage of the latent networking capabilities of the hardware used.

Problem

In 2018, Municipal Solid Waste (MSW) generation in the United States reached a record peak of 4.9 lbs. per person per day². That's 34.3 lbs. per week, 137.2 lbs. per month, 1788.5 lbs. per year. In a lifetime³, a person at this rate of production would create nearly 140,000 lbs. of trash. When multiplied by the number of people living in the US, you get an estimated several hundred million pounds of trash produced per year. Over 50% of MSW that is properly disposed of is never repurposed⁴. This waste is predominantly disposed of in landfills, posing an enormous problem for the environment. This non-reusable waste is responsible for 14.3% of all US greenhouse gas emissions as of 2021, equivalent to the output of roughly 23.1 million cars worth of methane⁵.

These emission statistics do not begin to account for the fossil fuel emissions from operating these landfills, nor do they encompass the chemical contamination that may seep out into the surrounding ecosystem. All this is simply waste that is known to have been collected and in some way dealt with, be that placement in landfills, combusted for energy, or reused in some way. It does not account for the millions of pounds of trash that end up in the ocean every year,

² "National Overview: Facts and Figures on Materials, Wastes and Recycling." EPA, 3 Dec. 2022, www.epa.gov/facts-and-figures-about-materials-waste-and-recycling/national-overview-facts-and-figures-materials.

³ https://datacommons.org/tools/timeline#&place=country/USA&statsVar=LifeExpectancy_Person

⁴ Bodamer, D. (2016, November 16). *14 charts from the EPA's latest MSW estimates*. Waste360. <https://www.waste360.com/waste-reduction/14-charts-epa-s-latest-msw-estimates>

⁵ Environmental Protection Agency. (2023, April 21). *Basic Information about Landfill Gas*. EPA. [https://www.epa.gov/lmop/basic-information-about-landfill-gas#:~:text=Methane%20Emissions%20from%20Landfills,-Note%3A%20All%20emission&text=Municipal%20solid%20waste%20\(MSW\)%20landfills,of%20these%20emissions%20in%202021](https://www.epa.gov/lmop/basic-information-about-landfill-gas#:~:text=Methane%20Emissions%20from%20Landfills,-Note%3A%20All%20emission&text=Municipal%20solid%20waste%20(MSW)%20landfills,of%20these%20emissions%20in%202021).

and the negative impact that trash has on both human and animal life⁶. This is not sustainable, and by 2025 the daily trash production is expected to reach 6 million tons per day⁷.

Solution

A large problem like this requires a multitiered solution with engagement at every level, from industrial operations that produce thousands of tons per day down to the person who drops a small plastic wrapper on the street. The true solution is to change the disposal habits of the general public. Our solution will address the lowest level, a single person's disposal practices. We believe that one of the biggest issues with the production of excess waste in households is that people are for the most part ignorant of exactly how much trash they are producing. By empowering individuals to visualize and track changes in their trash production, our device provides the timely and concrete feedback necessary for people to make impactful alterations to their personal habits.

This feedback can be secured through measuring trash by weight, regularly recording the amount produced. However as we discovered, this is a lot harder to do than might first be imagined. The most straightforward way would be to, when taking the trash out, place it on a bathroom scale and record the weight, then input it into a spreadsheet and formulate it into a graph. However, this process is long, tedious, and not at all something people want to go through when dealing with a trash can. Our idea was to effectively automate this process by creating a

⁶ *Marine plastic pollution*. IUCN. (2022, July 20).

<https://www.iucn.org/resources/issues-brief/marine-plastic-pollution#:~:text=Over%20300%20million%20tons%20of,waters%20to%20deep%2Dsea%20sediments>.

⁷ Magazine, S. (2013, October 30). *When will we hit peak garbage?*. Smithsonian.com.

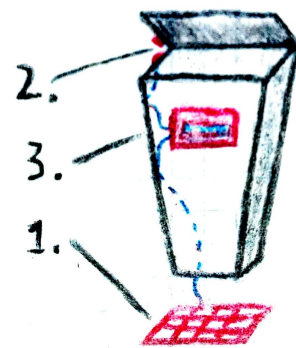
<https://www.smithsonianmag.com/science-nature/when-will-we-hit-peak-garbage-7074398/>

dedicated scale system that perpetually monitors and logs trash weight, storing it in a format that can be easily analyzed, manipulated, and viewed graphically in order to track your progress.

Prototype Design

Our project was designed from the start with the goals of being cheap and easy to manufacture, simple to set up and operate for the average user, and reliable over long periods of time. We needed it to be able to integrate with a wide range of trash cans and disposal systems; for example it should be able to fit a large kitchen trash can, but also should be able to support a small bathroom trash can if needed. Finally, the scale should log and display data correctly and in a form that is simple and easy to interpret.

From these criteria, we arrived at a prototype design with two main parts: The platform, and the control box. The platform is situated under the trash can, functioning in much the same way to a bathroom scale. It measures the total weight of the trash can using stress/strain sensors mounted on the bottom and sends this signal to the control box via a detachable cable. The control box houses the microcontroller, the SD card reader and RTC, the indicator light and buttons, as well as the LCD panel. This is where the signal from the platform as well as any inputs are processed, information about the trash can's current load as well as user prompts are displayed, and data is logged. The separation of these systems allowed us to modify the platform without endangering any electronic components, as well as to develop and test the control box without having to deal with the large and unwieldy platform. Another important factor in separating the control box from the platform



was the fact that the user interface is integrated with the control box. Joining this with the platform would not only be difficult but also position it on the ground, a location more susceptible to physical abuse and more difficult to access.

It is worth mentioning that we originally planned for a third part to this system: A sensor or switch package that would detect when the trash can was opened and closed, or when trash was deposited. The initial conception of the three parts of our system can be seen above in Figure 1. Label 1 indicates the platform, label 2 indicates the switch/lid sensor, label 3 indicates the control box. However, due to the immense variation in trash can openings and the resultant complexity this would necessitate, we chose to eschew the switch/lid sensor in favor of the simple albeit slightly less efficient method of constant, periodic weight measurements.

Development and Specifications

Throughout our development process, we used and maintained a 3D CAD model of our entire assembly using Autodesk Inventor 2023, as seen in Figure 2. The control box was programmed using C++ on PlatformIO with Microsoft Visual Studio Code. Various handheld, power, and CNC tools were used to machine and manufacture the parts that went into our assembly.

We decided to build our platform around a cheap plywood board mainly because it was a readily available material, it is easy to work with and CNC mill, and it suited our needs well. We cut it to 12" by 16" mostly arbitrarily, but this size was large enough to fit most any trash can yet small enough to work with on a tabletop. On the bottom of the platform we needed to fit the four stress/strain sensors wired in a Wheatstone bridge. The HX711 breakout board, which would

combine the signals from these sensors into a single signal readable by the control box, was to be installed on the side of the board in order to be as unobtrusive as possible.

The simple stress/strain sensors we ordered did not come with any mounts or hardware attachment points. Thanks to our access to a working FDM 3D printer, we were able to design and print plastic mounts that the sensors could clip into, with bolt holes for attachment to the board. Modeling plastic clips that would both hold the sensor firmly in place once installed yet not snap during assembly required some trial and error, but the final part worked as well as we could have hoped. The mounted sensors were fastened to the platform with 4M bolts,

countersunk on the top of the board to

maintain a flush surface for the trash can. The

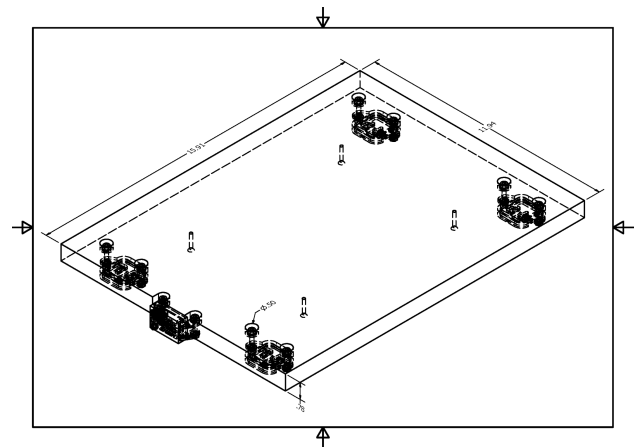
holes for these bolts were drilled precisely

using our lab's CNC router, then countersunk

with a drill press. The HX711 board was much

more difficult to install cleanly, largely

resulting from the configuration of its mount



holes and output pins as well as the fact that it was slightly wider than the thickness of the board.

The complex mount was manufactured using a resin 3D printer and took multiple attempts to

model and print it successfully. In the end the board fastened to the outside of the mount with a

pair of 3M bolts while the mount attached to the platform with a pair of 4M bolts through four

tabs. The countersink holes for the mount bolts were extended in a groove to the edge of the

board such that the mount could be slid on flush to the top of the platform. The slight overhang

of the mount on the underside of the platform provided a route for the sensor wires, which were

secured to the underside of the platform with hot glue.

The cable uses a four pin Dupont connector to connect, through an opening in the resin printed mount, to the pins in the HX711 board. On the other end it uses a four pin JST-XH connector to attach to the control box. We assembled the cable using ribbon wire threaded through a wire loom, with heat shrink and electrical tape reinforcing the ends near the connectors. Though functional, this cable is not very long and is weak at the joints between the wire and the connectors. We briefly considered using a networking cable for its solidity and EM shielding, but the required ports proved too large and cumbersome to fit into our design.

The control box was built around the Arduino Uno WiFi REV 2, the microcontroller which reads the various inputs and controls the outputs. This board was chosen for its widespread support, ease of use, reliability, as well as potential WiFi integration. The Arduino was paired with an Adafruit Data Logger Shield, a module which combines an SD card reader and RTC in addition to providing useful prototype board space. This functionality fit perfectly into our project which required accurate timestamps as well as the capacity to store large amounts of data in non-volatile memory, even when unable to connect to a network. The JST-XH connector for the platform, the RGB LED indicator light, the two input buttons, and the 20x4 character I²C LCD panel were all soldered to the prototype board area of the data logger shield. The indicator light, input buttons, and LCD panel were each attached with extended wires for added flexibility. The input buttons were installed on a separate prototype board due to limited space on the data logger shield, and so that the buttons could be installed in a more user-friendly position. We did not have enough time to design and print a case for the control box though the current prototype was designed with such a thing in mind. Each individual electronic component and circuit, as well as the whole system and its program, was tested on a breadboard before being soldered onto the prototype boards.

System Operation

The system is designed to be set up with the trash can placed on top of the platform. The control box may either be on the trash can/platform or off to the side, but this placement choice should be consistent over the course of each trash bag. The control box must have a working SD card inserted as well as supplied power via a USB-B cable in order to be fully initiated. Upon startup it will verify that each of its components is working and that the RTC has not lost time for some reason.

Logged data is stored in two separate files. One stores the current weight of trash, logged every time the weight has changed more than 0.02kg and has remained stable to within 0.01kg for more than 20 seconds. This ensures that the user is able to deposit multiple items, adjust or compact the bag, or even interfere with the platform without producing erroneous data. The second file logs the final weight of each trash bag before it is replaced. Trash bag removal is detected by large decreases in the measured weight of trash, confirmed by user input. When a trash bag is replaced, the weight of the scale is tared with the empty trash can before the new trash bag is inserted. This is a process that may also be initiated manually, for example when setting up the system for the first time. This duality of data logging methods allows for the analysis of both cumulative trash disposal, as well as when trash is deposited on a daily basis or over the course of a logged trash bag.

The user interface is rather simple, with prompts either requiring a simple confirmation or a yes/no input. Thus, only two buttons are necessary. The status of the system is indicated by a RGB LED, with red meaning that the system has yet to be fully initiated, yellow meaning that the weight is not stable or that there is a prompt the user must respond to, and green meaning that

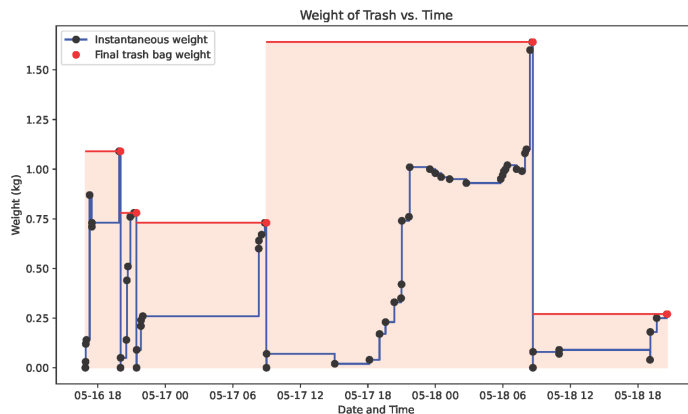
the weight is currently stable and that the system is operating nominally. The LCD panel has four rows, each with 20 characters, which provides more than enough room to display the necessary information. When waiting for a user response to a prompt, it will display the current date, time, and measured weight of trash.

Testing Process

Currently our prototype does not have any WiFi/IoT functionality, instead simply writing data to an SD card and relying on the user to analyze the raw CSV files that result. We have, however, written a simple Python script to read the data from the SD card and generate a graph of the gathered data. This was more than sufficient to test the operational functionality of our system and demonstrate what the visual representation of the gathered data might look like. However, due to limited testing time and lack of long-term functionality in our prototype, our testing could not determine whether or not our project might help an individual make changes in their waste disposal habits over time. The system was set up in one of our member's homes with a small dustbin for three days to collect the data seen below in the graph. Items were arbitrarily added to the trash can from a bag of paper waste in order to simulate trash disposal in a clean and more controlled manner. The trash can was periodically emptied back into the bag in order to log a new trash bag. Occasionally, an item would be removed from the trash can to simulate trash being removed from the bin.

Throughout this process, many flaws in the operational logic of the system, the user interface, and the physical assembly of certain components were identified. A loose connection on one of the buttons' leads caused false user input and otherwise interfered with the system's smooth operation. This was fixed, and the fragile joint was reinforced with hot glue. Due to the

lack of a protective casing around the prototype, joint weakness and total failure of connections is not to be unexpected. Feedback from the team member's family indicated that the incessant glow emitted from the LCD panel's backlight and the indicator light were aggravating, particularly at night. It proved difficult to design a well integrated solution to this issue during



the testing period. Finally, the system itself proved unreliable and difficult to operate at times, sometimes freezing and requiring a restart or logging changes where there should have been none. It is as yet

unclear how these issues will be resolved. However, this testing process successfully demonstrated that our system's weight logging processes will produce data that can be used to convey the amount of waste produced over time. According to the program that generated the plot in Figure 3, seen above, 4.51kg of waste were collected throughout the testing period.

Next Steps

While our system is functional at a basic level, it does not yet have the reliability or ease of use necessary for long term testing. This further testing will be paramount in determining how useful our system will be towards decreasing waste production over time in reality, as opposed to in theory. In the short term, there are many improvements that may be implemented without necessitating the construction of a new prototype. The simplicity and modularity of the Arduino's program must be improved, as this is a prerequisite to adding any new software

features such as a timeout for the LCD backlight/indicator light or menu options for resetting the system, etc. There remain occasional operational bugs which must also be remedied by these changes. More important than any software changes, however, is the necessity of an enclosure for the control box and its various components which currently remain loose, connected only by the wires. Aside from being a source of needless physical vulnerability, this has caused and will continue to cause issues with weakened or broken solder joints. Once these pressing issues have been solved, long term testing will be much more realistic.

When considering the improvements necessary to develop our prototype into a product that is deployable for indefinite use throughout a household, it is clear that implementing IoT networking capabilities will be critical. The most obvious benefit of this is that, with a well designed companion app, it would no longer be necessary to remove and read the SD card manually. All data would be offloaded from the microcontroller immediately to be stored and analyzed by the app. This presents another compelling avenue for simplification: If the device can be managed through an app, the cumbersome user interface on the control box is no longer necessary. With the help of a more compact microcontroller, it might then be possible to integrate the entire system into the platform, vastly improving the ease with which it can be set up as well as decreasing the cost and complexity of production. If these simplifications decrease the power draw of the system substantially enough, designing a model which runs on battery power might allow the system to be implemented in otherwise inconvenient locations. Once the platform has been simplified and its control has been centralized in an app, using multiple weighing platforms to monitor multiple trash receptacles throughout a household becomes the clear next step.

At this point, the applications for the data gathered by the system become limited only by what we can envision. This could be as simple as setting goals for your periodic trash production,

or as complex as comparing waste production in different rooms and creating tips based on the time of day that trash tends to be added in large amounts. Receptacles could be designated as recycling and compost such that the user can see positive change as well as negative change. The simplification of the user experience is key, as if it is not both easier and more effective, there is no advantage to using this product over repurposing a bathroom scale. Currently, our prototype is preferable to a cumbersome bathroom scale, but we would hope to add significantly more functionality before attempting to refine it and bring it to market.

Conclusion

Our testing shows that the current prototype works as designed, and with further improvement could be a viable solution to the issue of excess MSW production. Though the price point at which we would need to sell it to make a normal⁸ profit margin is still too high to be realistic, this is still a prototype. With a cheaper, simpler design that can be mass produced and includes IoT integration, the economy of scale could make it a viable product. While this solution may not be able to achieve a significant difference in the amount of trash produced every year at any conceivable scale of deployment, its strength lies in helping people develop strategies that they can use to reduce their own waste output as well as reinforcing mindfulness of their impact on the environment. The first step toward changing habits is to know them, and that is something our prototype is fully capable of enabling.

⁸ Based at 7-10% according to this article: Parker, T. (2022, October 30). *What's a good profit margin for a new business?* Investopedia. <https://www.investopedia.com/articles/personal-finance/093015/whats-good-profit-margin-new-business.asp#:~:text=But%20in%20general%2C%20a%20healthy,retail%20or%20food%2Drelated%20companies.>

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